

Bid Management & Vendor Coordination System

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I. Overview & Summary

This document provides a comprehensive implementation plan for an MVP Bid Management and Vendor Coordination System for BluOnX Development (land/horizontal) and Capstone LLC (vertical construction). The system streamlines the complete vendor lifecycle from bid invitation through contract award and milestone tracking, eliminating manual processes, ensuring standardized bid comparison, and preventing scheduling conflicts.

The implementation uses a custom secure form portal for bid submission with magic link authentication, providing complete control and a professional vendor experience. Post-award milestone tracking uses simple email-based check-ins to avoid overwhelming non-technical vendors. This MVP delivery will reduce bid cycle time from weeks to days, eliminate manual spreadsheet juggling, and provide complete audit trails for all vendor interactions.

Key Objectives:

- Automate vendor bid invitation and tracking with intelligent filtering
 - Implement secure magic link authentication for bid submission
 - Standardize bid collection through custom secure portal
 - Enable side-by-side bid comparison with intelligent scoring
 - Implement pre-award validation to prevent scheduling conflicts
 - Provide real-time dashboard visibility into bidding status
 - Generate award letters and contracts automatically with DocuSign integration
 - Implement simple email-based milestone tracking with automated deadline alerts
 - Ensure production-ready quality through comprehensive testing
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II. Tools & Technology Stack

Component	Technology	Purpose
Frontend	React 18 + Vite + TypeScript + TailwindCSS v4	Admin dashboard & UI
Backend	FastAPI + Python	API, business logic, vendor response handling
Database	PostgreSQL via Supabase	Data storage, RLS, triggers
Authentication	Supabase Auth (email/password only)	Internal user login
Bid Collection	Custom React Portal + Magic Links	Vendor bid submission
Email Service	AWS SES	Automated notifications, reminders, milestone check-ins
API Layer	React Query + Axios	Data fetching & caching
Testing	Playwright + Jest + pytest	Unit, integration, E2E testing
Automations	n8n Pro	Scheduled workflows, email orchestration, milestone check-in scheduling
Location Services	Google Maps API	Distance calculation, geocoding
E-Signatures	DocuSign API	Contract signing & execution
Hosting (Frontend)	Vercel Pro	Frontend deployment
Hosting (Backend)	AWS (EC2/ECS/Lambda)	Backend API hosting
Version Control	Git + GitHub	Code management

III. Real-Life Scenario

Imagine Samantha, a Project Manager at BluOnX Development, starting a new excavation and grading project:

1. Project Creation Samantha logs into the dashboard using her email and password and creates "Sunset Hills Phase 2" project with a \$2.5M budget, uploads site plans and specifications, and adds 8 tasks including excavation, utilities, and paving with individual budgets and required timelines.

2. Automated Bid Invitations She clicks "Start Bidding" on the Excavation & Grading task (\$145K budget). The system automatically filters 7 qualified excavation vendors within 75 miles using Google Maps distance calculation who have current insurance, adequate bonding, and available capacity. It generates unique secure portal links with magic link authentication for each vendor with pre-filled company information, and sends professional bid invitation emails with all project documents attached.

3. Real-Time Tracking Over the next week, the dashboard updates in real-time showing "3 of 7 vendors submitted." The system automatically sends friendly reminders at T-7 days, urgent reminders at T-3 days, and final call on the due date to non-responders via n8n automated workflows.

4. Intelligent Comparison On the bid deadline, Samantha opens the comparison view. The system displays all 3 bids side-by-side with complete line-item breakdowns. It normalizes different pricing formats (ABC uses lump sum, Johnson uses unit pricing), calculates weighted scores based on: 50% price, 5% compliance (insurance/docs), 20% past performance, 10% capacity, 15% timeline alignment. The system recommends ABC Excavation with a 94.5 score vs 89.2 for Johnson and 87.8 for XYZ.

5. Pre-Award Validation Samantha selects ABC Excavation and clicks "Award Contract." The system performs automatic validation: can start on required March 1st date, insurance certificate valid through project completion, currently 3 of 5 active jobs (has capacity), bid amount of \$145K within 5% of budget, bonding capacity adequate. All checks pass with no warnings.

6. Automated Award with DocuSign She confirms the award. Within seconds, the system: generates a detailed award letter with mobilization instructions and project milestones, sends congratulations email to ABC Excavation with DocuSign contract for e-signature, sends professional decline letters thanking the other 6 vendors for their submissions, updates ABC's capacity from 3/5 to 4/5 active jobs, creates the contract record with start/end dates, logs the entire transaction for audit trail.

7. Email-Based Milestone Tracking ABC Excavation's project starts with three milestones: "Site Clearing Complete" (March 8), "Rough Grading Complete" (March 15), and "Final Grade & Compaction" (March 22).

5 days before March 8: ABC receives automated email: "Reminder: Site Clearing work starts in 5 days"

March 8 (start date): ABC receives email with styled Yes/No buttons: "Did you start Site Clearing today?" ABC clicks [Yes, started on time].

March 10 (5 days before end): ABC receives email: "Will you complete Site Clearing on time by March 15?" ABC clicks [No, there's a delay].

Immediately: Samantha receives alert email: "ABC Excavation reported delay on Site Clearing. Contact: John Smith 555-1234." with a deep link into the dashboard milestone view where she can update the schedule.

Samantha calls ABC: Weather delay, need 3 more days. She clicks the email link, lands in the dashboard milestone detail view, updates end date to March 18. Automated checks restart with the new dates.

March 18: ABC receives completion check: "Did you complete Site Clearing?" ABC clicks [Yes, completed].

Immediately: Samantha receives notification: "ABC Excavation completed Site Clearing on Sunset Hills Phase 2."

The same automated cycle repeats for "Rough Grading Complete" and "Final Grade & Compaction" milestones.

8. Result: What used to take Samantha 2-3 weeks of manual work now takes 10 minutes. She has complete confidence that ABC Excavation can actually start on March 1st, complete audit trail for compliance, standardized bids for true apples-to-apples comparison, and all vendors were treated professionally. The \$145K excavation work is awarded to the best-qualified vendor with legally binding e-signature.

Throughout the project, automated email check-ins keep ABC accountable without overwhelming them with complex portal logins. Proactive alerts ensure Samantha stays ahead of any delays, eliminating the constant email chasing and manual status updates that used to consume hours of her day. She moves on to the next task with full visibility and control.

IV. Development Phases

Phase 1: Foundation & Database Setup (60 Hours)

Module 1: Initial Setup and Database Design

Task 1.1: Initialize Supabase project and configure environment - 3h

- Set up production and staging Supabase projects
- Configure environment variables and secrets management
- Tools: Supabase platform, environment setup

Task 1.2: Design and implement core database schema - 16h

- Tables: users, vendors, vendor_contacts, vendor_trades, vendor_documents, projects, project_documents, tasks, bid_templates, bid_template_items, bid_packages, bid_package_documents, bid_invitations, magic_link_tokens, bid_submissions, bid_line_items, bid_attachments, bid_scores, awards, contracts, docusign_envelopes, milestones, milestone_responses, milestone_alerts, email_log, vendor_flags, notifications, trades
- Relationships: foreign keys, junction tables, cascading rules
- Tools: PostgreSQL, Supabase database

Task 1.3: Create database indexes for query optimization - 4h

- Index vendor trade, location, status fields
- Index project and task status, dates
- Composite indexes for common query patterns
- Partial unique indexes on awards and contracts (one active per task)
- Tools: PostgreSQL

Task 1.4: Implement Row Level Security (RLS) policies - 4h

- Role-based access control (admin vs project_manager)
- Private schema helper functions (is_active_user, is_admin) with SECURITY DEFINER
- Read policies for all 28 tables (authenticated role)
- Write policies for admin-only tables (users, trades) and self-update tables (notifications, vendor_flags)
- Revoke all access for anon role
- Tools: Supabase RLS

Task 1.5: Set up database triggers - 8h

- Auto-update timestamps on all tables (fn_set_updated_at)
- Vendor capacity management (increment/decrement on award/contract status changes)
- Bid package round number auto-increment
- Bid invitation status sync on submission
- Denormalized field consistency enforcement (bid_submissions, awards, contracts, milestones)
- Auth user profile creation trigger
- Tools: PostgreSQL functions, PL/pgSQL

Task 1.6: Configure Supabase Storage buckets - 4h

- vendor-documents bucket (W-9, insurance certs, master trade agreements)
- project-documents bucket (plans, specs, photos)

- bid-attachments bucket (docs vendors upload with bids)
- RLS policies for secure access (authenticated = full CRUD, anon = zero access)
- Tools: Supabase Storage

Task 1.7: Set up authentication framework - 5h

- Configure Supabase Auth (email/password only)
- Auth context/provider with session and profile state
- Auth service layer (signUp, signIn, signOut, resetPassword, getAccessToken)
- ProtectedRoute component shell
- Session persistence and auto token refresh
- Tools: Supabase Auth, React

Task 1.8: Create development and staging environments - 6h

- Separate Supabase projects for dev/staging/prod
- Environment-specific configuration
- Database seeding scripts
- Tools: Supabase, environment config

Task 1.9: Configure AWS infrastructure - 10h

- Set up AWS account and IAM roles
- Configure EC2/ECS or Lambda for FastAPI backend
- Set up load balancer and auto-scaling
- Configure CloudWatch monitoring
- Request AWS SES production access (submit early to avoid delays)
- Tools: AWS Console, AWS CLI

Acceptance Criteria:

- Supabase project fully configured with production and staging environments
- Complete database schema with 28 tables, all relationships, and indexes
- RLS policies protect sensitive data appropriately
- Automated triggers handle capacity updates, timestamps, denormalized field consistency, and round numbering
- Authentication framework configured for email/password
- AWS infrastructure ready for backend deployment
- AWS SES production access requested
- Storage buckets configured with proper access controls

Phase 2: Frontend Foundation, Backend API & Authentication (100 Hours)

Module 2: React Application Setup, FastAPI Backend & Authentication

Task 2.1: Initialize React + Vite project with TailwindCSS - 4h

- Project scaffolding with Vite
- TailwindCSS v4 configuration and custom theme
- Tools: React 18, Vite, TailwindCSS v4, TypeScript

Task 2.2: Create reusable UI component library - 15h

- Buttons, inputs, forms, selects, checkboxes
- Cards, modals, dialogs, alerts
- Tables with sorting and filtering
- Loading states and skeletons
- Navigation components
- Data visualization components
- Form validation components
- Tools: React, TailwindCSS

Task 2.3: Build FastAPI backend - 25h

Detailed Breakdown:

- FastAPI project initialization and structure (folder structure, virtual environment, dependencies, configuration management)
- Pydantic v2 models for request/response validation
- Supabase admin client via lifespan context manager and dependency injection (single service-role client, not per-request)
- JWT validation middleware (PyJWT, HS256, audience="authenticated") for authenticating Supabase tokens
- API endpoint scaffolding for all major resources: vendors, projects, tasks, bids, contracts, DocuSign endpoints, milestones (RESTful route structure)
- CORS configuration for Vite frontend (localhost:5173 for dev, Vercel domain for production)
- Docker containerization for backend
- Tools: FastAPI, Python, Uvicorn, Pydantic v2, PyJWT, Docker

Task 2.4: Configure Supabase client for state management - 5h

- Supabase client initialization (singleton)
- Auth state management with onAuthStateChange (synchronous fire-and-forget pattern)
- Real-time subscription setup
- fetchProfile with AbortController timeout protection

- Tools: Supabase JavaScript client

Task 2.5: Set up React Query for API layer and caching - 6h

- React Query configuration
- Custom hooks for data fetching
- Cache invalidation strategies
- Tools: React Query, Axios

Task 2.6: Set up routing with React Router v6 - 6h

- Define route structure for all major pages
- Protected route components
- Navigation guards
- Tools: React Router v6

Task 2.7: Build authentication system - 25h

Detailed Breakdown:

- Email/password authentication flow (login, register, email verification, password reset, session management)
- Session management and persistence (token storage, automatic token refresh via Supabase client, logout and session cleanup)
- Protected route guards and RBAC (route protection HOCs, role-based UI rendering, permission checking utilities)
- Tools: Supabase Auth, React, FastAPI

Task 2.8: Create main dashboard layout with navigation - 8h

- Sidebar navigation with icons
- Top header with user menu
- Breadcrumb navigation
- Responsive hamburger menu for mobile
- Tools: React, TailwindCSS

Task 2.9: Implement responsive design - 6h

- Mobile (320px+), tablet (768px+), desktop (1024px+) breakpoints
- Touch-friendly UI elements
- Mobile navigation patterns
- Tools: TailwindCSS responsive utilities

Acceptance Criteria:

- React application running locally and deployable to Vercel Pro
- FastAPI backend running with Supabase connection via service-role key
- Clean, responsive UI with consistent design system across all screen sizes
- Email/password authentication fully functional with password reset
- Session persistence working correctly (users stay logged in)
- Protected routes redirect unauthenticated users to login
- FastAPI backend properly integrated with Supabase database
- CORS configured correctly for frontend domain
- API endpoints scaffolded and ready for business logic
- JWT validation middleware authenticating Supabase tokens correctly
- Authentication working end-to-end with proper error handling
- Routing structure in place for all major pages
- Reusable component library ready for rapid development

Phase 3: Core Entity Management (65 Hours)

Module 3: Projects, Tasks, Vendors, and Bid Templates CRUD

Task 3.1: Build vendor management interface - 15h

- Vendor list view with search, sort, filter
- Add/edit vendor form with validation
- CSV import with field mapping and validation
- Vendor detail page with history
- Delete vendor with confirmation (soft delete)
- Advanced filtering and bulk operations
- Tools: React, FastAPI, Supabase API, CSV parsing library

Task 3.2: Implement project management - 12h

- Project list view with status indicators
- Create project form with document upload
- Edit project details and budget tracking
- Project detail page with task list
- Budget allocation and tracking
- Project timeline visualization
- Tools: React, FastAPI, Supabase API

Task 3.3: Create task management within projects - 10h

- Add task form within project context (trade dropdown filtered by task phase)
- Edit and delete tasks (soft delete)

- Task dependencies and sequencing via sort_order
- Budget allocation per task
- Timeline and deadline setting
- Bid type selection (competitive, direct_assign, internal)
- Tools: React, FastAPI, Supabase API

Task 3.4: Build document upload functionality - 10h

- Drag-and-drop file upload component
- File type validation (PDF, JPEG, PNG, DWG)
- File size limits and validation
- Document preview and download
- Secure URL generation via Supabase Storage
- Coat check pattern: metadata in DB, files in storage buckets
- Tools: React, Supabase Storage, file handling libraries

Task 3.5: Integrate Google Maps for location - 8h

- Address geocoding for vendors and projects
- Distance calculation between vendor and project
- Map display for vendor locations
- Radius filtering visualization (75-mile default)
- Tools: Google Maps JavaScript API, Distance Matrix API

Task 3.6: Build bid template management interface - 10h

- Create/edit bid templates per trade (lump sum vs structured line items)
- Manage bid template items (description, item_type, unit_of_measure, sort_order)
- Associate templates with trades
- Preview template structure
- Tools: React, FastAPI, Supabase API

Note on budget validation: Task budget sum validation against project total budget is enforced at the FastAPI application layer, not at the database level.

Acceptance Criteria:

- Admin can import 100+ vendor records via CSV with field validation
- Complete CRUD operations for vendors, projects, tasks, and bid templates
- Project documents uploaded and accessible via secure links
- Data validation prevents invalid entries (negative budgets, past dates)
- Task budget sum validated against project total budget at API layer
- Google Maps integration calculates accurate distances within 75-mile radius
- Vendor filtering by radius works reliably
- Bid templates configurable per trade (lump sum or structured line items)

- All forms have proper error handling and user feedback

Phase 4: Bid Invitation System (50 Hours)

Module 4: Automated Vendor Filtering & Invitation

Task 4.1: Build intelligent vendor filtering algorithm - 10h

- Trade/specialty matching with multiple trade support
- Geographic radius calculation using Google Maps Distance Matrix API (75-mile default)
- Insurance validity date checking (block expired vendors)
- Bonding capacity verification against project value
- Current capacity check (active jobs vs maximum)
- Vendor rating/history filtering
- Display vendor flags as warnings during selection
- Tools: PostgreSQL queries, FastAPI, Google Maps API

Task 4.2: Create bid invitation email templates - 6h

- Professional HTML email design
- Responsive email layout (mobile-friendly)
- Dynamic content placeholders (vendor name, project details)
- Tools: HTML/CSS email templates

Task 4.3: Implement AWS SES email service integration - 8h

- Configure AWS SES account and verify domain
- SPF, DKIM, DMARC records setup
- API authentication and credentials
- Email sending function with attachments
- Set up AWS SNS topic for bounce/delivery notifications
- Bounce notification webhook (FastAPI or n8n endpoint)
- Update email_log with sent/failed/bounced status
- Tools: AWS SES SDK, AWS SNS

Task 4.4: Build batch email sending function - 8h

- n8n workflow for batch email processing
- Rate limiting to respect SES service limits
- Retry logic for failed sends
- Progress tracking and status updates
- Error logging to email_log table
- Email personalization engine
- Tools: n8n Pro, AWS SES

Task 4.5: Implement invitation tracking - 6h

- Invitation status tracking (sent, opened, submitted, declined, expired, no_response)
- Submission status updates
- Dashboard status indicators
- Tools: Database updates, n8n workflows

Task 4.6: Build vendor selection UI - 12h

- Filterable vendor list for task (auto-filtered then PM-reviewed)
- Bulk select/deselect vendors
- Preview filtered vendors before sending
- Manual vendor addition/removal (PM override)
- PM selects which project documents to include via bid_package_documents
- Send invitation confirmation dialog
- Tools: React, UI components

Acceptance Criteria:

- System accurately filters vendors by all criteria (trade, distance within 75 miles, insurance, bonding, capacity)
- Google Maps API calculates distances within 75-mile radius accurately
- Professional bid invitation emails render correctly across email clients
- Unique secure portal links with magic link authentication generated per vendor
- Selected project documents attached to invitation emails successfully
- Batch email sending respects SES rate limits with proper queuing via n8n
- Invitation status tracked in real-time on admin dashboard
- Email failures logged and retried automatically
- Bounce notifications received via SNS and recorded in email_log
- PM can preview and override vendor list before sending invitations
- Vendor flags displayed as warnings during vendor selection

Phase 5: Bid Collection - Custom Secure Forms (60 Hours)

Module 5: Custom Bid Portal with Magic Link Authentication

Task 5.1: Design custom bid submission UI - 12h

- Multi-step form design (company info, pricing, documents, review)
- Progress indicator component
- Responsive mobile layout
- Form field validation UX
- File upload interface
- Draft save functionality

- Enhanced accessibility features
- Polish and user experience refinements
- Tools: React, TailwindCSS, UI/UX design

Task 5.2: Build magic link authentication system - 12h

Detailed Breakdown:

- Token generation: secure random token generation, SHA-256 hashing (raw token emailed, only hash stored in magic_link_tokens table)
- Email delivery: HTML email template with styled magic link button, link formatting and deep linking, resend link functionality
- Token validation middleware in FastAPI: hash lookup in magic_link_tokens, expiration checking (24-48 hours), vendor identity validation, rate limiting on token requests
- Session management for vendors: stateless JWT issued after successful magic link validation (short-lived, 2-4 hours), no server-side session store needed. If JWT expires, vendor clicks magic link again and resumes from their saved draft.
- Security hardening: IP address logging, rate limiting per vendor, suspicious activity detection
- Tools: FastAPI, cryptography libraries, PyJWT

Task 5.3: Implement form backend API - 8h

- Create bid submission endpoints
- Multi-part form data handling
- Real-time validation during entry
- Draft saving endpoint
- File upload handling to bid-attachments bucket
- Final submission endpoint
- Tools: FastAPI, Supabase, file handling

Task 5.4: Build form validation logic - 6h

- Client-side validation (React Hook Form)
- Server-side validation (Pydantic v2 models)
- Pricing format validation (lump sum, unit pricing)
- Bid template pre-population (check if task's trade has a template; if yes, show structured line items; if no, show lump sum field)
- Required field enforcement
- Business logic validation (dates, amounts)
- Custom validation rules per field
- Tools: React Hook Form, Pydantic v2, FastAPI

Task 5.5: Implement file upload for vendors - 6h

- Drag-and-drop file upload
- Multiple file support (including signed Scope of Work per bid)
- File type restrictions (PDF, JPEG, PNG)
- File size limits and file-type validation at API layer
- Upload progress indicators
- Tools: React Dropzone, Supabase Storage, FastAPI file validation

Task 5.6: Build draft save functionality - 6h

- Auto-save draft every 2 minutes (bid_submissions with is_draft = true)
- Manual save button
- Draft retrieval on page reload (via vendor JWT session)
- Draft expiration after deadline
- Progress preservation
- Tools: React, FastAPI, Supabase

Task 5.7: Create submission confirmation flow - 6h

- Review page showing all entered data
- Edit buttons to go back to sections
- Final submission button with confirmation
- Submission success page
- Confirmation email generation
- PDF receipt generation
- Tools: React, email templates, PDF generation library

Task 5.8: Implement vendor portal routing - 4h

- Public routes for magic link landing
- Form access validation
- Redirect logic for expired tokens
- Submission status page
- Error pages (404, 403, expired)
- Tools: React Router

Acceptance Criteria:

- Magic links work reliably (99%+ success rate)
- Tokens expire correctly after 24-48 hours or after submission
- Vendor cannot access form with expired token
- Stateless JWT session allows form interaction without re-validating token on every request

- Multi-step form saves progress automatically
- Real-time validation provides instant feedback
- Vendors can upload signed Scope of Work, insurance certificates, and supporting documents
- Draft functionality allows vendors to return and continue (JWT re-issued via magic link click)
- Mobile experience is smooth on phones and tablets
- Submission confirmation sent immediately via email
- Form accessible only via magic link (no direct URL access)
- Security testing validates token cannot be reused or forged
- Bid template line items pre-populated when trade has a defined template
- Submission PDF receipt generated for vendor records

Phase 6: Real-Time Dashboard & Response Tracking (60 Hours)

Module 6: Live Bid Status Tracking

Task 6.1: Build real-time submission tracking - 12h

- Supabase real-time subscriptions setup
- WebSocket connection management
- Submission count live updates ("3 of 7 submitted")
- Status change notifications
- Connection health monitoring
- Fallback to polling if WebSocket fails
- Tools: Supabase Realtime, React, WebSocket

Task 6.2: Create bid tracking dashboard - 18h

- Overview cards (total invited, submitted, pending)
- Vendor list with submission status indicators
- Timeline view of bid activity
- Submission timestamp display
- Quick actions (resend invitation, view bid)
- Real-time progress bars
- Tools: React, Chart.js, UI components

Task 6.3: Implement completeness checker UI - 12h

- Visual indicators for complete/incomplete bids
- Missing items list per vendor
- Color-coded status badges
- Completeness percentage
- Action buttons (contact vendor, mark complete)

- Tools: React, validation logic

Task 6.4: Build basic charts and visualizations - 18h

- Bar chart comparing bid amounts
- Pie chart of submission status distribution
- Timeline chart of submission activity
- Score comparison radar chart
- Interactive tooltips and data exploration
- Export visualizations functionality
- Tools: Chart.js, Recharts, data visualization

Acceptance Criteria:

- Dashboard updates within 5 seconds of submission (via WebSocket)
- Submission count displays accurately in real-time
- Incomplete bids flagged with specific missing items
- Charts update automatically without page refresh
- Vendor list shows clear status indicators (pending, submitted, incomplete)
- Admin can take actions directly from dashboard
- WebSocket connection resilient with automatic reconnection
- Performance remains smooth with 50+ concurrent bids
- No manual refresh needed for any real-time data

Phase 7: Automated Reminder & Alert System (44 Hours)

Module 7: Deadline Reminders, Document Expiration Monitoring & Escalation

Task 7.1: Create bid reminder scheduling logic in n8n - 8h

- n8n workflow for reminder scheduling
- Date calculation logic (T-7, T-3, T-0 before bid deadline)
- Timezone handling and conversion
- Vendor status checking (exclude vendors who already submitted)
- Batch processing for multiple projects
- Tools: n8n Pro, date libraries

Task 7.2: Build tiered email templates - 6h

- Friendly reminder template (7 days before deadline)
- Urgent reminder template (3 days before deadline)
- Final call template (due date)

- Personalized content per vendor
- All templates as professional HTML with styled buttons/links
- Tools: HTML email templates, dynamic content

Task 7.3: Implement reminder sending workflow - 8h

- n8n workflow execution via daily cron
- Email batch sending through AWS SES
- Status tracking (reminder sent timestamps) logged to email_log
- Error handling and logging
- Bounce notification processing
- Tools: n8n Pro, AWS SES

Task 7.4: Create admin escalation alerts - 6h

- Daily summary email to admins
- List of non-responding vendors
- Overdue submission alerts
- Recommended actions
- Dashboard notification badges
- Tools: Email templates, n8n Pro

Task 7.5: Build reminder history log - 6h

- Display reminder send history per vendor
- Admin view of all reminders sent (via email_log)
- Resend reminder manually option
- Tools: React, database queries

Task 7.6: Build document expiration monitoring workflow - 10h

- n8n daily cron workflow: query vendor_documents.expiration_date and vendors.insurance_expiration_date
- Send reminder emails at T-30 days and T-7 days before expiration
- HTML email template with vendor name, document type, and expiration date
- Record all alerts in email_log
- Dashboard indicator for vendors with expiring/expired documents
- Tools: n8n Pro, Supabase nodes, AWS SES

Acceptance Criteria:

- Bid reminders automatically sent at T-7, T-3, and T-0 intervals
- Only non-submitting vendors receive bid reminders (submitted excluded)
- Escalating urgency in email tone (friendly, urgent, final)
- n8n workflows run reliably with monitoring

- Admin receives escalation email after deadline with missing vendor list
- Reminder history logged in email_log table for audit
- System handles timezone differences correctly
- Manual resend option available from admin dashboard
- Reminder send failures logged and retried
- Document expiration reminders sent at T-30 and T-7 days
- Expired vendor documents flagged in dashboard

Phase 8: Bid Comparison & Scoring (40 Hours)

Module 8: Intelligent Bid Analysis

Task 8.1: Build bid normalization engine - 10h

- Parse different pricing formats (lump sum, unit pricing, mixed)
- Convert all bids to comparable totals
- Handle missing or incomplete pricing data
- Currency and decimal handling
- Line-item breakdown normalization
- Tools: JavaScript/TypeScript, FastAPI, business logic

Task 8.2: Implement weighted scoring algorithm - 10h

Scoring Weights (client-confirmed):

- 50% Price (inverse curve, lowest price scores highest)
- 5% Compliance (insurance validity, required documents, onboarding status)
- 20% Past Performance (milestone on-time completion rate + vendor flag history; new vendors get neutral/baseline score)
- 10% Capacity (available jobs vs maximum)
- 15% Timeline Alignment (can start on required date, schedule fit)

Implementation details:

- Edge case handling (no history, missing data, new vendors)
- bid_scores.scored_by: NULL = system-generated, non-NULL = manually adjusted by PM
- Scoring metadata snapshot stored in bid_scores.scoring_metadata (JSONB)
- Tools: Python scoring logic, PostgreSQL, FastAPI

Task 8.3: Create side-by-side comparison UI - 12h

- Responsive table with all bids
- Sortable columns (price, score, timeline)

- Color-coded indicators (green/yellow/red)
- Expandable line-item details
- Highlight recommended vendor (highest score)
- Filter and search functionality
- Enhanced mobile-responsive design
- Print-friendly view
- Tools: React, data tables library, UI components

Task 8.4: Build recommendation engine - 8h

- Automatic highest-scoring vendor identification (recommendation only, never auto-award)
- Justification report generation
- Alternative recommendations (2nd, 3rd place)
- Warning flags (over budget, late start, missing docs, expired insurance)
- Override mechanism for PM (with justification required)
- Tools: Business logic, report generation

Acceptance Criteria:

- All bids normalized to comparable format regardless of pricing structure
- Weighted scores calculated accurately on 0-100 scale using client-confirmed weights
- Scoring handles edge cases (new vendors, incomplete data) gracefully
- Comparison table displays all bids with complete line-item breakdowns
- Highest-scoring vendor recommended with clear reasoning (PM always makes final decision)
- PM can sort by any column (price, score, timeline, compliance)
- Color-coded warnings show issues (over budget, conflicts, missing docs)
- UI is polished, professional, and easy to understand
- Export comparison to Excel/PDF for client sharing
- PM can manually adjust individual scores with justification

Phase 9: Award Decision & Contract Generation (40 Hours)

Module 9: Pre-Award Validation & Automation

Task 9.1: Build comprehensive pre-award validation - 8h

- Start date feasibility check against vendor schedule
- Insurance validity through project completion date (block if expired)
- Bonding capacity validation against project value
- Vendor capacity check (not at maximum jobs)
- Budget variance check (within +/- 5% of estimate)
- Prior performance validation

- Conflict detection (scheduling, resources)
- Tools: Business logic, PostgreSQL, FastAPI, validation engine

Task 9.2: Implement override workflow - 4h

- Warning dialog with specific conflict details
- Required justification text field (stored in awards.override_justification)
- awards.has_override flag set to TRUE
- Validation results snapshot stored in awards.validation_results (JSONB)
- Override history logging for audit trail
- Tools: React forms, workflow logic, database

Task 9.3: Integrate DocuSign API - 10h

- DocuSign account setup and API credentials
- Contract template creation in DocuSign
- Envelope creation and sending
- Signer configuration (vendor, owner)
- Webhook for signature completion (updates contract status and docusign_envelopes)
- Document retrieval after signing
- Tools: DocuSign API, FastAPI, webhook handling

Task 9.4: Create award letter email template - 3h

- Professional congratulations message (HTML email)
- Project details and expectations
- Mobilization instructions and timeline
- Next steps and contact information
- DocuSign link embedded
- Tools: HTML email template, dynamic content

Task 9.5: Build contract record creation - 8h

- Generate contract record in database (one per accepted award)
- Auto-generate contract_number (unique)
- Store start date, end date, contract amount, payment terms
- Link DocuSign envelope ID via docusign_envelopes table
- Update vendor capacity via existing database triggers (award accepted = +1)
- Handle direct_assign flow: create synthetic bid_submission (is_direct_assign = TRUE) so downstream chain works identically
- Tools: FastAPI, Supabase, database triggers

Task 9.6: Implement decline notification system - 7h

- Professional decline letter template (HTML email)

- Thank you message for participation
- Feedback option for future improvements
- Batch sending to all non-selected vendors via n8n
- Maintain positive vendor relationships
- Tools: n8n Pro, AWS SES, email templates

Acceptance Criteria:

- All validation checks execute automatically before award
- Validation flags: start date conflicts, expired insurance, insufficient bonding, over capacity, budget variance
- Warning dialogs show specific conflicts with details
- PM can override warnings with documented justification (logged for audit)
- DocuSign envelope created and sent automatically upon award
- Award letter email sent with DocuSign link for e-signature
- Professional decline letters sent to all non-selected vendors within minutes
- Vendor capacity automatically updated via database trigger
- Contract record created with complete details (dates, amount, terms)
- Direct assign flow creates synthetic submission and follows same pipeline
- Complete audit trail maintained for compliance
- DocuSign webhook updates contract status when signed

Phase 10: Vendor Commitment & Performance Tracking (48 Hours)

Module 10: Email-Based Milestone Tracking & Automated Alerts

Overview: Simple email-driven milestone tracking that does not overwhelm vendors. No complex portal. Automated check-ins via styled HTML emails with Yes/No button links, and alerts to PM when intervention is needed.

Architecture:

- Outbound (scheduling, tokens, emails): n8n owns entirely. Reads milestones from Supabase, generates tokens, stores hashes directly in Supabase via Supabase nodes, sends HTML emails through AWS SES.
- Inbound (vendor click responses): FastAPI owns entirely. Validates tokens, records responses, updates milestone status, returns styled HTML confirmation pages, fires webhook to n8n for follow-up.
- Follow-up (PM alerts, notifications): n8n handles, triggered by FastAPI webhook.
- PM actions (date updates, manual overrides): React dashboard via deep links from PM alert emails.

Task 10.1: Build milestone management interface - 12h Admin interface for creating and managing milestones:

- Add milestones to awarded tasks (name, start date, end date). Milestones are optional and PM-controlled.
- Milestone list view with color-coded status on project/task detail page (gray: scheduled, blue: started, green: on_track, yellow: delayed, red: no response, checkmark: completed)
- Edit and delete milestone functionality
- PM manual override capabilities: "Mark as started" with manual actual_start_date, "Mark as completed" with manual actual_end_date
- Date update form (accessible via deep link from PM alert emails): update end date, reset status to on_track, restart automated check cycle
- Notes field for PM to add context about delays or issues
- Milestone detail view showing full response history from milestone_responses
- Tools: React, Supabase API

Task 10.2: Build email response link handlers - 10h Backend endpoints for vendor and PM email interactions:

- FastAPI endpoint: /v1/milestones/respond/{token}?response=yes|no
- Token validation: SHA-256 hash lookup in milestone_alerts.response_token_hash, check expiration (7 days), check milestone is still active
- First-response-wins logic: first click records the official response in milestone_responses. Subsequent clicks (including email scanner pre-fetches or accidental double-clicks) show a friendly confirmation page: "You already responded [Yes] on [date]"
- Response recording in database with timestamp and vendor_contact_id
- Automatic milestone status updates based on response (yes on start = started, yes on completion = completed, no = delayed)
- Set actual_start_date/actual_end_date on appropriate responses
- Styled HTML confirmation pages returned to vendor (no login required)
- Async webhook to n8n after recording response (triggers follow-up workflows)
- No-response escalation: if vendor has not responded within 2 days, n8n triggers PM alert (once per check-in, not repeatedly)
- Tools: FastAPI, token generation, HTML templates

Task 10.3: Build n8n automated alert workflows - 12h Three core automation workflows, all using Supabase nodes for direct DB access:

Daily Schedule Checker workflow (runs 8 AM daily):

- Check milestones 5 days before start date (only if milestone duration > 7 days): send reminder to vendor + PM

- Check milestones on start date: send start confirmation request to vendor with Yes/No buttons
- Check milestones 5 days before end date (only if milestone duration > 7 days): send progress check to vendor with Yes/No buttons
- Check milestones on end date: send completion confirmation to vendor with Yes/No buttons
- Check milestones with no response for 2+ days: escalate to PM (once per check-in)
- Generate secure token per email, hash with SHA-256, store hash in milestone_alerts via Supabase node
- Embed raw token in email button links pointing to FastAPI:
<https://api.bluonx.com/v1/milestones/respond/{token}?response=yes|no>
- Log all alerts sent in email_log via Supabase node

Vendor Response Handler workflow (triggered by FastAPI webhook):

- Process "No" responses from vendors
- Update milestone status to "delayed" (if not already handled by FastAPI)
- Send immediate alert to PM with vendor contact details and deep link to dashboard milestone view
- Log alert in email_log

Completion Notification workflow (triggered by FastAPI webhook):

- Process completion confirmations from vendors
- Send notification to PM that work is finished
- Log completion notification in email_log

Delay re-check cycle: after a vendor responds "No", the automated cycle pauses for that check type. PM must update the end date (which resets status to on_track and restarts the cycle with new dates) or manually mark the milestone as complete from the dashboard.

- Tools: n8n Pro, Supabase nodes, webhook triggers, AWS SES

Task 10.4: Create email templates - 5h Seven professional HTML email templates with styled Yes/No buttons:

- Starting Soon email (to vendor + PM): 5 days before start (only for milestones > 7 days duration)
- Start Confirmation email (to vendor): on start date with Yes/No buttons linking to FastAPI
- Progress Check email (to vendor): 5 days before end (only for milestones > 7 days duration) with Yes/No buttons
- Completion Confirmation email (to vendor): on end date with Yes/No buttons
- Delay Alert email (to PM): when vendor reports delay, includes vendor contact info and deep link to dashboard milestone view for date update
- No Response Alert email (to PM): when vendor does not respond for 2+ days

- Completion Notification email (to PM): when vendor confirms completion
- All templates branded with BluOnX branding
- Dynamic content placeholders for project, task, vendor, and date details
- Mobile-responsive HTML design
- Tools: HTML email templates, dynamic content system

Task 10.5: Build PM dashboard widgets - 7h Dashboard components for milestone visibility:

- Milestone Timeline View: visual timeline with color-coded status (gray: scheduled, blue: started, green: on_track, yellow: delayed, red: no response, checkmark: completed)
- Active Milestones List with filtering and sorting, quick status overview, "Update Date" button for delayed milestones (links to milestone detail view), click to view response history
- Alerts/Notifications Panel showing recent activity (delays, no responses, completions), click to view details, mark as reviewed functionality
- Integration with main dashboard for quick overview
- Tools: React, Chart.js for timeline visualization, UI components

Task 10.6: Implement vendor flagging system - 5h Manual PM flagging for problematic vendors:

- Flag button on vendor detail page
- Reason dropdown: Missed deadline, Poor quality, Unresponsive, Other
- Notes field for PM to add context
- Optional link to specific milestone if flag is related to milestone issue (vendor_flags.milestone_id)
- Flag display on vendor record with date and reason
- Warning icon in vendor lists for flagged vendors
- Flag warnings displayed during vendor selection for new bids (in Phase 4 filtering UI)
- Resolve/unflag functionality when issues resolved (only flag creator or admin can resolve)
- Tools: React, Supabase API

Acceptance Criteria:

- PM can create milestones with start/end dates for awarded tasks (milestones are optional)
- Automated HTML emails sent at correct times:
 - T-5 days before start: reminder to vendor + PM (only if milestone duration > 7 days)
 - On start date: start confirmation to vendor with Yes/No buttons
 - T-5 days before end: progress check to vendor with Yes/No buttons (only if milestone duration > 7 days)

- On end date: completion check to vendor with Yes/No buttons
- Vendor email responses work reliably:
 - Styled [Yes] [No] buttons link to FastAPI endpoint
 - First click records official response; subsequent clicks show "already responded" page
 - Simple styled confirmation page shown after click (no login/portal required)
 - Email scanner pre-fetches and double-clicks handled gracefully
- PM gets alerted when needed:
 - Vendor clicks [No]: PM gets email with vendor contact info and dashboard deep link
 - No response for 2 days: PM gets escalation email (once per check-in)
 - Work completed: PM gets notification email
- PM can update end dates:
 - Deep link from alert email lands in dashboard milestone detail view (authenticated)
 - Status resets to on_track after date update
 - Automated checks restart with new dates
- PM can manually override milestone status:
 - "Mark as started" with manual actual_start_date
 - "Mark as completed" with manual actual_end_date
- Delay cycle pauses after "No" response until PM updates date or marks complete
- Dashboard shows milestone status with timeline view, active list, and alerts panel
- PM can manually flag problematic vendors with reason, notes, and optional milestone link
- Flags display as warnings in vendor lists and bid invitation flow
- All vendor communications logged in email_log for audit trail
- All vendor responses preserved in milestone_responses as immutable records

Phase 11: Comprehensive Testing & Quality Assurance (80 Hours)

Module 11: Thorough Testing Across All Layers

Task 11.1: Unit Testing - 20h

- Vendor filtering algorithm with edge cases (no vendors in radius, all expired insurance, etc.)
- Bid normalization logic (various pricing formats)
- Scoring algorithm with different scenarios and client-confirmed weights
- Date calculations for reminders and deadlines
- Database trigger testing (capacity management, denormalized field consistency, round numbering)
- Validation functions (insurance expiration check, bonding, dates)

- Magic link token generation and SHA-256 hashing validation
- Milestone check-in date calculations (including short milestone duration handling)
- Email response link token validation (first-response-wins logic)
- Code coverage target: >80% for critical business logic
- Tools: Jest, pytest, test fixtures

Task 11.2: Integration Testing - 20h

- Custom Vendor Portal: magic link click, JWT session, form fill, draft save, final submission full pipeline
- Email delivery pipeline (SES send, SNS bounce notification, email_log update)
- Authentication flows (email/password login, session refresh, logout)
- File upload/download to Supabase Storage (all three buckets)
- Real-time sync reliability (WebSocket subscriptions)
- Database trigger chains (award accepted, capacity update, contract creation)
- n8n workflow execution (bid reminders, milestone check-ins, document expiration alerts)
- API error handling and retry logic
- FastAPI backend with frontend integration
- DocuSign envelope creation and webhook processing
- Google Maps distance calculation accuracy (75-mile radius)
- Milestone response pipeline: email button click, FastAPI token validation, response recording, n8n webhook trigger, PM alert email
- Milestone date update: dashboard deep link, date change, status reset, n8n cycle restart
- Document expiration monitoring: n8n detection, email send, email_log recording
- Tools: Postman, automated testing, test data

Task 11.3: End-to-End Testing - 20h

- Complete bid cycle: create project, add task, configure bid template, invite vendors, receive bids, compare, award, DocuSign
- Complete milestone cycle: create milestone, automated check-ins, vendor response via email button, PM alert, date update via dashboard, completion
- Direct assign flow: PM selects vendor, synthetic submission created, award, contract, milestones
- Internal task flow: budget line only, no bid/award/contract/milestone
- Multi-user scenarios (concurrent admin actions)
- Mobile responsive testing (real devices)
- Performance testing (dashboard load time, real-time updates)
- Error scenarios (network failures, timeouts, invalid data, expired tokens)
- Magic link access and form submission
- Email button clicks and response recording (including double-click and scanner pre-fetch scenarios)
- Tools: Playwright, Cypress, real device testing

Task 11.4: User Acceptance Testing (UAT) - 10h

- Test with 5-7 real test vendors
- Client walkthrough of complete workflows
- Test milestone email responses with real vendors (actual email button clicks)
- Test document expiration reminder emails
- Feedback collection and prioritization
- Edge case discovery from real-world usage
- Documentation of issues and feature requests
- Refinement based on client feedback
- Tools: Real vendor data, client sessions

Task 11.5: Bug Fixes & Performance Optimization - 10h

- Critical bug resolution from testing phases
- Performance optimization (query optimization, caching)
- Error handling improvements
- UI/UX polish based on feedback
- Final regression testing
- Load testing verification
- Tools: Debugging, profiling, optimization

Acceptance Criteria:

- All critical functions have passing unit tests (>80% coverage)
- Integration tests pass for all major workflows
- Complete bid cycle tested successfully end-to-end with 5-7 test vendors
- Complete milestone tracking tested with real email responses (button clicks, not just API calls)
- Direct assign and internal task flows tested
- System tested on mobile (iOS/Android), tablet, and desktop
- Email/password authentication tested thoroughly
- Magic link authentication tested thoroughly (including expiration and reuse scenarios)
- Milestone first-response-wins logic tested (double-click, scanner pre-fetch)
- n8n workflows tested for reliability (bid reminders, milestone check-ins, document expiration)
- PM dashboard deep links tested end-to-end
- FastAPI backend thoroughly tested with all endpoints functional
- DocuSign integration tested with test envelopes
- Google Maps integration tested for 75-mile distance accuracy
- Known bugs documented, prioritized, and critical issues resolved
- Edge cases handled gracefully with user-friendly error messages
- Performance targets met: dashboard < 2 seconds, real-time updates < 5 seconds
- Client confirms system meets requirements through UAT

- No critical or high-priority bugs remaining

Phase 12: Deployment & Training (60 Hours)

Module 12: Production Launch & User Enablement

Task 12.1: Configure production environment - 10h

- Deploy backend to AWS (EC2/ECS/Lambda with load balancer)
- Deploy frontend to Vercel Pro
- Configure custom domain and SSL certificates
- Configure production environment variables
- Tools: AWS Console, Vercel Pro dashboard, DNS configuration

Task 12.2: Set up production Supabase - 8h

- Production database configuration
- Automated backup schedules (daily, weekly)
- Performance monitoring and alerts
- Tools: Supabase Pro dashboard

Task 12.3: Configure production email service - 6h

- AWS SES production credentials and domain verification
- Confirm SPF, DKIM, DMARC records are active and valid
- SNS bounce/delivery notification topics connected to production endpoints
- Test email deliverability to multiple providers (Gmail, Outlook, corporate)
- Tools: AWS SES, DNS management

Task 12.4: Set up production n8n workflows - 8h

- Production n8n instance configuration
- Deploy all workflows: bid reminders, milestone check-ins, document expiration monitoring, escalation alerts
- Production Supabase and AWS SES credentials configured in n8n
- FastAPI webhook URLs updated to production endpoints
- Monitoring and error alerting
- Tools: n8n Pro

Task 12.5: Configure monitoring and alerting - 8h

- AWS CloudWatch for backend monitoring
- Vercel Analytics for frontend
- Supabase monitoring dashboard

- Error tracking (Sentry or equivalent)
- Uptime monitoring (UptimeRobot)
- Alert notifications (email)
- n8n workflow execution monitoring
- Tools: CloudWatch, Vercel Analytics, Sentry, UptimeRobot

Task 12.6: Create admin documentation - 8h

- User manual with annotated screenshots
- Workflow guides for each major process
- Troubleshooting section
- FAQ document
- Tools: Documentation software, screenshots

Task 12.7: Create video walkthrough tutorials - 6h

- Recording of major workflows (15-20 minutes total)
- Creating a project and inviting vendors
- Comparing bids and making awards
- Managing vendors and documents
- Using the dashboard
- Creating and tracking milestones
- Tools: Screen recording, video editing

Task 12.8: Conduct live training session - 6h

- Screen sharing demonstration
- Live Q&A with client team
- Hands-on practice session
- Feedback collection
- Tools: Zoom, screen sharing

Acceptance Criteria:

- System deployed and accessible via custom domain
- Frontend on Vercel Pro with optimized performance
- Backend on AWS with auto-scaling and load balancing
- Production environment stable and secure
- Automated daily backups configured and tested
- Email deliverability verified with SPF/DKIM/DMARC records
- SNS bounce handling active in production
- All n8n workflows deployed, configured, and monitored in production
- Monitoring and alerting active for errors and performance
- Client can successfully perform all operations independently

- Documentation covers all major workflows with screenshots
- Video tutorials available for onboarding new team members
- Client confirms understanding and satisfaction with system
- Post-launch support plan established

V. Estimated Hours & Timelines

Phase	Module	Key Tasks	Hours
Phase 1	Foundation & Database	Database schema (28 tables), RLS, triggers, storage, auth framework, AWS setup	60
Phase 2	Frontend + Backend + Auth	React + TailwindCSS, FastAPI on AWS, email/password auth	100
Phase 3	Entity Management	Vendor/project/task CRUD, bid template UI, Google Maps	65
Phase 4	Bid Invitations	Vendor filtering (75mi), AWS SES integration, batch sending via n8n	50
Phase 5	Bid Collection	Custom portal + magic link auth + stateless JWT sessions	60
Phase 6	Real-Time Dashboard	WebSocket tracking, charts, live updates	60
Phase 7	Reminder & Alert System	n8n bid reminders (T-7, T-3, T-0), document expiration monitoring (T-30, T-7)	44
Phase 8	Comparison & Scoring	Bid normalization, weighted scoring (50/5/20/10/15), comparison UI	40
Phase 9	Awards & Contracts	Pre-award validation, DocuSign integration, direct assign flow	40
Phase 10	Milestone Tracking	Email-based check-ins (n8n outbound, FastAPI inbound), PM dashboard, vendor flagging	48
Phase 11	Comprehensive Testing	Unit, integration, E2E, UAT, bug fixes	80
Phase 12	Deployment & Training	AWS/Vercel/n8n deployment, SES production, docs, training	60

TOTAL			707
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Estimated Development Hours: ~707 hours (range: 565-850 hours for MVP scope) Estimated Timeline: 11 weeks (range: 9-13 weeks for MVP scope)

VI. Estimated Costing

1. Infrastructure Costing (Monthly)

Service	Plan	Cost/Month	Notes
Supabase	Pro	\$25	Database, auth, storage, 100GB bandwidth
Vercel	Pro	\$20	Frontend hosting, serverless functions, analytics
AWS (Backend)	EC2/ECS	\$100 (estimated)	t3.medium or Fargate, load balancer, varies by traffic
AWS SES	Pay-as-you-go	~\$1-5	\$0.10 per 1,000 emails. At MVP scale (under 1,000 emails/month) cost is minimal
AWS SNS	Pay-as-you-go	~\$1	Bounce/delivery notifications from SES
n8n	Pro	\$59	Automations, workflows, 300 workflows
Google Maps API	Pay-as-you-go	\$40	Distance Matrix, Geocoding, ~10k requests
DocuSign	Standard	\$30	10 envelopes/user/month, e-signatures
Domain	Annual	\$12	Custom domain (.com)
SSL Certificate	Included	\$0	Provided by Vercel/AWS
Monitoring	Sentry (Team)	\$29	Error tracking, performance monitoring
Backups	AWS S3	\$15	Automated backup storage

Total (Per-Month)		~\$325/mo (+/- 20%)	For complete deployed system
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VII. Additional Notes

Assumptions

- Client provides a complete vendor list with all required data fields (name, trade, location, contact, insurance dates, bonding capacity)
- Project documents are in standard formats (PDF, JPEG, PNG, DWG)
- AWS account with appropriate IAM permissions for infrastructure setup
- DocuSign account for e-signature integration
- Client team available for requirements clarification, UAT testing, and training sessions
- Initial vendor count: 100-200 vendors across all trades
- Typical project: 5-10 tasks requiring bids, 5-7 vendors per task
- Testing includes 5-7 willing test vendors for the UAT phase
- Milestone tracking uses email-only approach based on client preference
- Vendors are not highly technical; simplicity is critical
- Post-MVP features (advanced analytics, photo uploads for milestones, vendor self-onboarding) would be developed after MVP success

Dependencies

- Client weekly meetings to vet the scope or help scoping for the sprint/module
- Timely access to Supabase account credentials and project setup
- AWS account with admin access for infrastructure provisioning (including SES production access request submitted early)
- AWS SES domain verification and SPF/DKIM/DMARC DNS records
- Google Maps API key with Distance Matrix and Geocoding enabled
- DocuSign API credentials and account setup
- n8n Pro account for workflow automations
- Client approval on the database schema design before Phase 1 completion
- Sample project data and documents for testing (provided by Week 2)
- 5-7 test vendors willing to receive bid invitations during UAT
- Test vendors willing to respond to milestone check-in emails during UAT
- Custom domain for production deployment
- DNS access for email configuration and domain setup
- Vercel Pro and AWS accounts for hosting

Out of Scope (MVP)

The following features are explicitly NOT included in this MVP and will be developed in future versions:

- Mobile native apps (iOS/Android)
 - Advanced analytics and reporting dashboards
 - Automated payment processing and invoicing
 - Multi-company/multi-user role management with granular permissions
 - Change order management
 - Subcontractor management and tier-2 vendor coordination
 - Budget forecasting and variance analysis
 - SMS notifications via Twilio
 - Weather delay tracking
 - Equipment scheduling and management
 - Photo uploads for milestone completion
 - Complex vendor portal for milestone tracking (keeping it email-based)
 - Google Sign-In / OAuth authentication
 - Vendor self-onboarding portal
 - Contingency budget tracking with variance workflow
 - Email open tracking and click tracking (email_log.opened_at and clicked_at deferred; action tracking via dedicated tables is sufficient for MVP)
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VIII. Conclusion

This ~707-hour MVP delivers a production-ready Bid Management & Vendor Coordination System that will transform BluOnX Development and Capstone LLC's vendor workflow.

The custom secure form portal approach for bid submission provides a professional vendor experience with complete control, while the simplified email-based milestone tracking respects the technical limitations of construction vendors and avoids overwhelming them with complex portal workflows.

This implementation delivers:

- Elimination of manual spreadsheet juggling and email chains
- Secure magic link authentication for vendors with stateless JWT sessions
- Robust FastAPI backend hosted on AWS infrastructure for performance and scalability
- Professional frontend deployment on Vercel Pro with optimized delivery
- Intelligent vendor filtering using Google Maps distance calculation (75-mile radius)
- Automated bid collection with real-time tracking through a custom portal
- Objective, data-driven vendor selection through weighted scoring (50% price, 5% compliance, 20% performance, 10% capacity, 15% timeline)
- Pre-award validation to ensure vendors can actually perform on required timelines

- Legally binding contract execution through DocuSign integration
- Simple, email-based milestone tracking with styled HTML emails and one-click Yes/No buttons
- Hybrid architecture: n8n for outbound scheduling and email orchestration, FastAPI for inbound vendor response handling
- Automated check-ins that prevent delays without overwhelming vendors
- PM alerts only when human intervention is needed, with deep links into the dashboard
- Document expiration monitoring with automated reminders at T-30 and T-7 days
- Complete audit trails for compliance and decision documentation
- Comprehensive testing ensuring production-ready quality
- Professional email automation via AWS SES with bounce tracking via SNS
- Workflow automation through n8n Pro for reminders, milestone check-ins, and alerts

Key Advantages of This Approach:

For Vendors:

- Simple email-based communication (no complex portals to learn)
- One-click responses to check-ins via styled buttons
- No need to log in repeatedly
- Professional, respectful treatment

For Project Managers:

- Automated tracking without manual follow-up
- Alerts only when intervention needed
- Dashboard deep links from alert emails for quick action
- Complete visibility into milestone status

For the Business:

- Earlier identification of delays
- Better vendor accountability
- Complete audit trail
- Professional, consistent communication
- Proactive document expiration management

Total Hours: ~707 hours Timeline: 11 weeks Monthly Infrastructure Cost: ~\$325 ROI Timeline: 6 months

Appendix: Changes from Previous Plan (v2.0 to v2.1)

This section summarizes all updates and decisions made during the v2.1 review. Items are grouped by category.

Removed from MVP

1. **Google Sign-In / OAuth:** Removed from all phases and acceptance criteria. MVP uses email/password authentication only via Supabase Auth.
2. **Redis:** Removed from tech stack. Email queueing handled by n8n Pro. Vendor bid portal sessions handled by stateless JWT issued after magic link validation.
3. **ClamAV virus scanning:** Removed. File-type and file-size validation at the FastAPI API layer is sufficient for MVP.
4. **Email open and click tracking:** Deferred to post-MVP. The email_log schema columns remain but will not be populated. Action tracking via dedicated tables (bid_submissions, milestone_responses) is sufficient.
5. **Supabase Edge Functions:** Removed from Task 4.1. All server-side logic routes through FastAPI.

Corrected from Client Feedback

6. **Scoring weights updated:** Changed from 40/20/20/10/10 to client-confirmed weights: 50% price, 5% compliance, 20% past performance, 10% capacity, 15% timeline alignment.
7. **Distance filter updated:** Changed from 50 miles to 75 miles per client request.
8. **Award process:** Explicitly documented as 100% manual. System recommends but never auto-awards.
9. **Milestones are optional:** PM-controlled, not required for every task. Short-duration tasks can skip milestones or use a single completion milestone.

Added to MVP

10. **Bid template management UI:** Added as Task 3.6. Admin interface for creating/editing bid templates per trade (lump sum vs structured line items). This was identified as a gap after the original plan.
11. **Document expiration monitoring:** Added as Task 7.6. n8n daily workflow sends reminder emails at T-30 and T-7 days before vendor document expiration. Client specifically requested this during requirements review.
12. **PM manual override for milestones:** Added to Task 10.1. PM can manually "Mark as started" or "Mark as completed" with date entry, handling cases where vendors respond outside of the email flow.

Architecture Decisions

13. **Email service: AWS SES confirmed.** Cheaper, already in AWS ecosystem, sufficient for MVP needs. Bounce tracking via AWS SNS webhooks. SES production access to be requested during Phase 1.9.
14. **Milestone tracking architecture (Hybrid Approach C):** n8n owns entire outbound side (scheduling, token generation, token hash storage directly to Supabase, email sending via SES). FastAPI owns entire inbound side (vendor click handling, token validation, response recording, HTML confirmation pages). n8n handles follow-ups via FastAPI webhook triggers. PM actions route through the React dashboard.
15. **Milestone token design:** Tokens are time-limited (7 days) with first-response-wins logic, not single-use. This prevents issues with email scanner pre-fetches and accidental double-clicks.
16. **Short milestone handling:** T-5 day reminder emails only sent if milestone duration exceeds 7 days. Start/end date check-ins always sent regardless of duration.
17. **Delay re-check cycle:** After a vendor responds "No", the automated check cycle pauses. PM must update the end date (resets to on_track, restarts cycle) or manually mark the milestone complete.
18. **PM date-update flow:** PM alert emails include deep links into the React dashboard milestone detail view. PM is already authenticated via session. No separate mini-app or standalone form needed.
19. **n8n confirmed as essential:** Handles all time-triggered outbound workflows (bid reminders, milestone check-ins, document expiration monitoring, escalation alerts). FastAPI handles all request-driven inbound logic.

Plan Cleanup

20. **Task 4.4 removed:** Was an empty placeholder ("Custom form setup - See Phase 5"). Phase 4 tasks renumbered.
21. **Task 10.2 (milestone schema) removed:** Milestone database tables were already built in Phase 1 (Tasks 1.2, 1.3, 1.4, 1.5). Hours reallocated to enhanced milestone response handling.
22. **Phase 3 hour breakdowns added:** Per-task hours now specified (were previously only a phase total).
23. **Phase 2 task order updated:** Reordered to reflect logical execution sequence (2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9).
24. **"Phase 2 features" terminology:** References to post-MVP features now say "Post-MVP" to avoid confusion with development Phase 2.
25. **Budget validation noted:** Task budget sum validation against project total is enforced at FastAPI application layer, not at the database level.
26. **Infrastructure cost updated:** AWS SES replaces SendGrid in cost estimate. Redis removed. Monthly estimate adjusted to ~\$325.

